

ATHUL PK

Full Stack Developer

MongoDb, Express.js, React.js, Node.js

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SUMMARY

A self-trained **full stack developer** who has experience designing responsive web applications using the **React.js, MongoDB, Express.js, Node.js** (MERN stack). Experienced at designing User Interfaces as well as understanding how they work, **HTTP** and **API** design and integration is also an area of proficiency. In terms of rapid development, my expertise lies in using **Backend as a Service** (BaaS) type platforms, including **Firebase** and **Supabase** for speedier authentication, real-time databases and **cloud computing** (AWS, Vercel, etc..) implementations. I have a solid background in security fundamentals, essential command line tools and **Git/GitHub**. Additionally, I am well versed in performance and scalability optimization, basic knowledge about **CI/CD pipelines**, and **System design**. I am looking forward to working as a junior **Frontend** and **Backend** developer while continuing to focus on developing secure, scalable, and user centered solutions.

TECHNICAL SKILLS

- **Languages & Core Web:** JavaScript (ES6+) , TypeScript , SQL , HTML5/CSS3, HTTP/HTTPS, API Design & Integration.
 - **Frontend Development:** React.js , Next.js , Redux Toolkit , Zustand , Vite, Responsive Web Design.
 - **Backend & Systems:** Node.js , Express.js , RESTful APIs , MVC Architecture, System Design Basics, Scalability Fundamentals
 - **BaaS & Cloud Platforms:** Supabase , Firebase, AWS (EC2, S3) , Vercel , Netlify, Render
 - **Databases:** MongoDB, PostgreSQL, MySQL
 - **Real-time & Payments:** Socket.IO, Razorpay, Stripe.
 - **Security & Authentication:** JWT (Auth & Auth) , Security Fundamentals.
 - **Tools & Version Control:** Git/GitHub, Command Line Tools (Bash), Postman, Figma, Basics of CI/CD Pipelines, Jest, React Testing Library.
 - **UI & Animation:** Tailwind CSS, Material UI (MUI), GSAP, Framer Motion, Styled Components
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PROJECTS

AI-LEARNING ASSISTANT – Full Stack

- Developed a **MERN**-based AI learning platform with **JWT** authentication, PDF upload/storage, and in-app document viewing.
- Integrated **Google Gemini AI** for contextual chat, summaries, explanations, flashcards, and automated quiz generation.
- Built interactive quizzes, progress analytics dashboard, and a modern **responsive UI** using **Tailwind CSS** for seamless cross-device use.

VIDEO EDITING WEB – Full Stack (Working on it)

- Engineered a browser-native editing suite using **FFmpeg.wasm** and **WebGPU**, enabling 4K video processing and hardware-accelerated real-time previews directly in the client browser.
- Developed a multi-track "Magnetic Timeline" using **Next.js** and **Zustand**, managing a highly nested global state for layers, keyframes, and real-time synchronization of audio-visual assets.
- Integrated AI segmentation models for automated background removal and speech-to-text APIs for Auto-Captioning, significantly reducing manual editing time for end-users.
- Architected a JSON-based project schema in **MongoDB** to track non-destructive edits and implemented a client-side rendering pipeline to export high-definition media without incurring server-side processing costs.

AUTO TAXI FINDER APP – Full Stack

- Developed a cross-platform mobile application using **React Native (Expo)** and **TypeScript** that enables passengers to visualize nearby auto-rickshaws within a 5km radius in real-time.
- Architected a robust backend using **Supabase**, leveraging **PostgreSQL** for data management and Real-time Engine for instantaneous driver location updates and availability status.
- Implemented fine-grained data access control using **Supabase** Row Level Security (RLS) to ensure driver and passenger data privacy, alongside secure **JWT**-based Authentication.
- Designed a high-aesthetic, **performance-focused UI** with deep localization for the Kerala market, focusing on **Performance and Scalability** to handle concurrent location streams with minimal latency.

REAL TIME CHAT WEB– Full Stack

- Engineered a high-performance communication layer using **Socket.io**, enabling instantaneous messaging, typing indicators, read receipts, and a collaborative synchronized whiteboard for developer workflows.
- Implemented a robust **MVC** (Model-View-Controller) backend with **Node.js** and **Express**, coupled with a frontend built on **React** and Context API for efficient global state management of auth and socket streams.
- Designed sophisticated **MongoDB** schemas to manage nested relationships for public/private channels, direct messages, and specialized features like "soft-deletes" and multi-user message reactions.
- Orchestrated a distributed deployment strategy using **Vercel** (Frontend) and **Render** (Backend), with **MongoDB Atlas** for cloud data persistence, ensuring a scalable and live-access production environment.

FOOD DELIVERY APP – Full Stack

- Built a **React Native (Expo)** food delivery app powered by a **MERN** stack backend with secure **JWT**-based authentication and structured **MongoDB** schemas.
- Developed **REST APIs** for core functionalities such as restaurant browsing, search/filter, cart, order placement, order history, and menu management (**CRUD**).
- Implemented real-time order tracking, reusable component architecture, and performance optimizations for a smooth user experience.

EDUCATION

MERN Stack Development	
Ash Academy	Aug 2025 -Jan 2026
High Secondary	
GHSS Puthupadi	March 2019 - March 2022

CERTIFICATION

<u>Artificial intelligence</u>
UL Research & OSPF
<u>Red Team Academy</u>
Certified Cyber Security Analyst
<u>Frontend Development</u>
Oneroadmap
<u>DevOps Engineer</u>
Oneroadmap
<u>React.JS Developer</u>
Oneroadmap

INTERESTS

Designing,Editing,Music,Video Gaming,Exploring New Tech,Photography and Videograpghy